

VIZUALX

On-Device Multimodal AI Perception & Intelligent Navigation for the Visually Impaired

Project Dossier & Whitepaper | Architecture, Speech Gating, Phonetic Normalization & Multimodal AI

Executive Summary

VizualX is an on-device, continuous multimodal AI perception assistant engineered for visually impaired users. Unlike conventional computer vision tools that act as raw sensor stream announcers, VizualX was designed around a fundamental human principle: **Audio bandwidth is cognitive bandwidth**. The application pairs high-speed local computer vision reflexes with an intelligent **Silence-by-Default AI Speech Arbiter**, a **Phonetic Speech Normalizer**, an on-demand **Assistive Reading Engine**, and a hybrid **Local Gemma 2B / Google Gemini 1.5 Flash Cloud Vision** intelligence stack.

Core Engineering Rule (Silence is a Valid State): *The system monitors 3D space continuously into internal memory, but strictly silences spoken output unless an immediate, actionable safety hazard is detected or the user explicitly queries the environment.*

1. The Problem Space & The Flaws of Current Assistive Tech

Assistive mobility applications frequently fail in real-world situations because they are built as raw object detectors rather than actionable assistants. Testing revealed five critical pain points:

- **Audio & Sensory Fatigue (The Chatterbox Failure):** Traditional tools speak every detected object in view ('Bench on left... Tree ahead... Chair... Person... Sign'). This creates severe cognitive overload and dangerously masks real environmental acoustic cues like traffic sounds, echoes, and footfalls.
- **Hallucinated Directions & Non-Actionable Speech:** Describing objects as 'on your left' or 'on your right' without millimeter head-pose tracking is inaccurate. Saying 'Caution, vehicle' fails to tell a blind user what physical action to take (Stop? Step back? Proceed?).
- **Robotic 'Letter-by-Letter' TTS Spelling:** Stock Android Text-to-Speech engines treat ALL-CAPS words (e.g. CAUTION, EXIT, STOP, ROOM 101) as acronyms, spelling them out as 'C-A-U-T-I-O-N' rather than pronouncing fluent spoken words.
- **Cloud Latency & Network Fragility:** Cloud-only vision APIs produce 2–5 second latency and blackout in subways, basements, and transit corridors where safety hazards require sub-100ms response times.
- **Visual UI Clutter:** Visual buttons and menus on mobile screens are inaccessible to blind users who need an intuitive, full-screen gesture surface.

2. The Solution: What We Built Across VizualX

VizualX solves these challenges through a modular, decoupled multimodal architecture combining local on-device perception, strict speech arbiters, and generative visual reasoning:

A. Silence-by-Default AI Speech Arbiter (ContextEngine)

Sensors monitor objects, text, and acoustic volume at 5 FPS into internal world state memory, but the **Speech Arbiter** enforces strict gating:

- **Critical Safety Hazards (Spoken Immediately):** Approaching vehicles in the walking path, descending stairs/drop-offs, collision obstacles closing within 1.8m in the center trajectory, emergency sirens, and hazard warning signs (e.g. 'Caution Wet Floor').
- **Ambient Background Clutter (100% Silent):** Benches, trees, peripheral pedestrians walking past, and non-hazard signs (storefronts, room numbers) are tracked silently in memory without triggering audio speech.

- **Global Quiet Buffer:** A mandatory 5.0-second silence interval between routine informational messages ensures continuous quiet and situational awareness.

B. Phonetic Speech Normalizer & OCR Synthesizer (SpeechNormalizer)

To eliminate robotic letter-by-letter spelling, every spoken string is processed through a dedicated phonetic normalizer before audio synthesis:

- **De-Acronymization:** Converts all-caps text (*CAUTION, RESTROOM, EXIT*) to sentence case so the TTS engine pronounces natural words while preserving genuine acronyms (*ATM, GPS, SOS, VIP*).
- **Spaced Character Collapsing:** Merges fragmented camera OCR characters ('*S T O P*' → '*Stop*', '*E X I T*' → '*Exit*').
- **Spatial & Building Abbreviation Expansion:** Automatically expands '*St.*' to '*Street*', '*Rm.*' to '*Room*', '*Bldg.*' to '*Building*', '*No.*' to '*Number*', and '\$*12.50*' to '*12 dollars and 50 cents*'.

C. On-Demand Assistive Reading Tool (AssistiveReadingEngine)

A dedicated tool allowing visually impaired users to read documents, mail, food packaging, door plaques, and menus on demand:

- **Spatial Text Flow:** ML Kit OCR text blocks are sorted top-to-bottom and left-to-right into coherent human reading paragraphs.
- **Dual Trigger Modes:** Triggered instantly by a single tap anywhere on the full-screen camera or via voice query ('*Read this*', '*What does this document say?*').

D. Hybrid Intelligence: Local Gemma 2B & Google Gemini 1.5 Flash

- **On-Device LLM Brain (LocalLLmBrain):** Uses quantized Gemma 2B / Phi-3 via MediaPipe LLM Inference to synthesize immediate active directive action commands under 8 words ('*Stop immediately, vehicle moving ahead*', '*Obstacle close, sweep cane*') without network connection.
- **Google Gemini 1.5 Flash Cloud Vision (GeminiApiClient):** Provides multimodal visual reasoning for complex queries ('*Describe what is in front of me*', '*Is there an open seat?*', '*What store is across the hallway?*').

E. Full-Screen Edge-to-Edge Gesture UX (MainScreen)

The user interface has zero buttons or visual clutter. The entire display is a 100% full-screen camera view with clean top tab pills for **Camera Feed** and **Google Maps Nav**. Blind accessibility is powered by universal touch gestures:

- **Single Tap:** Triggers Assistive Reading on the current camera frame.
- **Double Tap:** Activates hands-free Voice AI inquiry.
- **Long Press:** Toggles Pause / Resume perception.

3. Technical Architecture & Capability Comparison

Capability / Metric	Traditional Assistive Tools	VizualX Intelligent System
Speech Gating Policy	Continuous chatter; describes every object, tree, and bench.	Silence is Valid ; speaks only for imminent safety hazards or user queries.
Sign & Text Handling	Spells uppercase words letter-by-letter (C-A-U-T-I-O-N); OCR noise.	Phonetic Speech Normalizer ; speaks fluent words; hazard signs alerted; ambient signs read on demand.
Hazard Latency	1500ms – 4000ms (Cloud API roundtrip latency).	< 50ms on-device reflex pipeline (MediaPipe + heuristic arbiter).
Navigation Output	Vague spatial labels ('Person on left').	Direct Action Verbs ('Stop immediately', 'Sweep cane').
Generative Intelligence	None or cloud-only text summary.	Hybrid : On-device Gemma 2B LLM + Google Gemini 1.5 Flash Multimodal Vision.
Interface Design	Visual UI with buttons and small tap targets.	100% Full-Screen Gesture UX (Single tap, double tap, long press).

4. Verification & Testing Matrix

The codebase has been verified with **39 comprehensive automated unit tests** covering the entire pipeline:

- **Context & Speech Arbiter Tests:** Verified critical vehicle warnings, stair drop-offs, side-object suppression, and hazard vs. ambient sign speech gating.
- **Phonetic Normalizer Tests:** Verified de-acronymization of all-caps signs, spaced letter merging (*S T O P* → *Stop*), abbreviation expansions, and OCR noise removal.
- **Assistive Reading Tests:** Verified spatial text sorting and OCR paragraph extraction lifecycle.
- **Local LLM Brain Tests:** Verified prompt templates, action command sanitization, and deterministic fallback behavior.

5. Future Roadmap & Smart Glasses Form Factor

Because VizualX decouples the perception and world state engines completely from the visual UI, the entire core intelligence stack is ready for deployment directly onto **Smart Glasses** equipped with front-facing camera frames and bone-conduction audio transducers. Next iterations will incorporate stereo depth sensors, ultrasonic rangefinders, and edge NPU acceleration for next-generation accessibility.